

Finite Abelian II

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Problem 7.

Proof. Let R be a PID. $\forall a, b \in \mathbb{R}$, by definition of a PID, $(a, b) = (g)$ where $g = xa + yb$ where $x, y \in \mathbb{R}$. Similarly by the definition of a PID, $\exists h \in R \ni (x, y) = h$ thus $\exists u, v \in \mathbb{R} \ni x = hu$ and $y = hv$. Thus $g = xa + yb = hua + hvb = h(ua + vb)$. $k := ua + vb$ and $k \in (a, b) = (g)$, thus $g|k$. However $g = hk \implies k|g$. Thus $g = k \implies h = 1$. Thus $(x, y) = (1)$. Thus $\exists p, q \in R, \ni xq + yp = 1$. Thus $\begin{bmatrix} x & -p \\ y & q \end{bmatrix}$ is invertible with determinant 1. Moreover, $\begin{bmatrix} a & b \\ * & * \end{bmatrix} \begin{bmatrix} x & -p \\ y & q \end{bmatrix} = \begin{bmatrix} xa + yb & * \\ * & * \end{bmatrix} = \begin{bmatrix} g & * \\ * & * \end{bmatrix}$

Suppose (a) is maximal among the set of ideals generated by any element of RAC , for any possible invertible R and C . Suppose a is in the $(1, 1)$ entry of RAC .

For the i -th entry of the first row. Since we are in a PID, there is a Division Algorithm, so $a_{1,i} = qa_{1,1} + r$. Apply the column operation, $c_i - qc_1 \rightarrow c_i$. Thus the $0 < a_{1,i} = r < a_{1,1}$, but since a is the maximal ideal, $r = 0$. Thus $a_{1,1} | a_{1,i}$.

For the i -th entry of the first column. Since we are in a PID, there is a Division Algorithm, so $a_{i,1} = qa_{1,1} + r$. Apply the row operation, $r_i - qr_1 \rightarrow r_i$. Thus the $0 < a_{i,1} = r < a_{1,1}$, but since a is the maximal ideal, $r = 0$. Thus $a_{1,1} | a_{i,1}$.

Now apply the argument in 4. □

Problem 8.

Proof. Define the function $\deg : \mathbb{F}[X] \rightarrow \mathbb{Z}^+$ which $\forall k \in \mathbb{Z}^+, \forall f_1, \dots, f_k \in \mathbb{F}$, takes $\sum_{i=0}^k f_i x^i \rightarrow k$. $\forall g, h \in \mathbb{F}[X]$, $g = \sum_{i=0}^{k_1} a_i x^i$ and $h = \sum_{i=0}^{k_2} b_i x^i$. Thus $gh = \left(\sum_{i=0}^{k_1} a_i x^i \right) \left(\sum_{i=0}^{k_2} b_i x^i \right) = \sum_{i=0}^{k_1+k_2} c_i x^i$. Thus $\deg(g) = k_1, \deg(h) = k_2, \deg(gh) = k_1 + k_2$. Thus $\deg(h) \leq \deg(gh) \geq \deg(g)$.

$\forall n, d \in \mathbb{F}[X] \ni d \neq 0$, suppose $\deg(n) < \deg(d)$ take $q = 0$ and $r = n$.

Otherwise $\deg(n) \geq \deg(d)$,

let $n(x) = \sum_{k=0}^{\deg(n)} a_k x^k$ and $d(x) = \sum_{k=0}^{\deg(d)} b_k x^k$.

We can subtract from n , a suitable multiple of d such that it eliminates the highest degree term of n .

So $r(x) = n(x) - \frac{a_{\deg(n)} x^{d-n}}{b_{\deg(d)}} m(x)$. $\deg(r) < \deg(f)$. If $\deg(r) \geq \deg(d)$ continue the process.

Eventually, $r(x) = n(x) - d(x) * \left(\frac{a_{\deg(n)}}{b_{\deg(d)}} x^{n-d} + \dots \right)$

Thus $\deg(r) < \deg(d)$ and $q(x) = \left(\frac{a_{\deg(n)}}{b_{\deg(d)}} x^{n-d} + \dots \right)$.

Thus $\mathbb{F}[X]$ has a division Algorithm and $\deg$ is a euclidean function.

Thus $\mathbb{F}[X]$ is a euclidean domain.

$F \subset F[X]$ naturally. Let M be a S -module. M is an abelian group which F acts on. Thus M is a F -vector space. However $f(x) \in F[X]$ also acts on M . So one must define a function from $\phi : S \rightarrow \text{Hom}_F(M, M)$ which maps $f(x)$ to some linear map.

$\forall f(x) \in F[X], f(x) = \sum_{k=0}^n f_k x^k, \forall k, f_k \rightarrow f_k I$, or scalar multiplication, because F acts by scalar multiplication on any F vector space. Thus $\phi(f(x)) = \sum_{k=0}^n f_k I \circ \phi(x^k)$

Moreover X^k is the repeated multiplication of X in $F[X]$ and must map to the repeated composition of $\phi(X)$ in $\text{Hom}_F(M, M)$ (ie $\phi(X) \circ \dots \circ \phi(X)$) since ϕ is a ring homomorphism. Thus $\phi(X^k)$ is completely determined by where $\phi(X)$ goes. Thus $\phi(X)$ uniquely determines ϕ . □

Problem 9.

Proof. $\implies$: Suppose M is a S -module and $\exists f \in F[X]/\{0\} \ni \forall m \in M, fm = 0 \implies f(A) = 0$. By Classification of S modules, $M = \bigoplus_{k=1}^n \left(\frac{F[X]}{(p_k)} \right) \oplus F[X]^r$. $\forall k, \left(\frac{F[X]}{(p_k)} \right) \cong \text{span}(1, x, \dots, x^{\deg(p_k)-1})$, thus $\left(\frac{F[X]}{(p_k)} \right)$ is finite dimensional and $\bigoplus_{k=1}^n \left(\frac{F[X]}{(p_k)} \right)$ is finite dimensional.

Assume the contrary, $r \neq 0$. Then take $\forall q_1, \dots, q_r \in F[X]/\{0\}$, then $m = (0, \dots, 0, q_1(x), \dots, q_r(x)) \in M$ and $f(x)(0, \dots, 0, q_1(x), \dots, q_r(x)) = (0, \dots, 0, f(x)q_1(x), \dots, f(x)q_r(x)) = f(A)m = 0$ so $\forall i, f(x)q_i(x) = 0$. But since $F[X]$ is an integral domain, $f(x)q_i(x) \neq 0$ since $f(x) \neq 0$ and $q_i(x) \neq 0$.

Thus $r = 0$ by contradiction. Thus $M = \bigoplus_{k=1}^n \left(\frac{F[X]}{(p_k)} \right)$ and thus is finite dimensional.

$\Leftarrow$: Suppose M is a finite dimensional F vector space and a S -module. Thus $M \cong \bigoplus_{k=1}^n \left(\frac{F[X]}{(p_k)} \right)$.

Take $f = \prod_{k=1}^n p_k$.

$\forall q \in F[X], fq \in (p_i)$ for all i . Thus $\forall m \in M, fm = 0$, because each element of m will be in the ideal of (p_i) and thus will be 0. □

Problem 10.

Proof. $\implies$: Suppose $F^n = M_A \cong M_B = F^n$ as S modules. Thus there $\exists$ a S -linear isomorphism $\alpha : M_A \rightarrow M_B$, such that $\forall m \in M_A, \forall f(X) \in F[X] \phi(f(A)m) = f(B)\phi(m)$. Thus the following commutative diagram is observed,

$$\begin{array}{ccc} F^n & \xleftarrow{\alpha} \twoheadrightarrow & F^n \\ \downarrow A & & \downarrow B \\ F^n & \xleftarrow{\alpha} \twoheadrightarrow & F^n \end{array}$$

Thus $B = \alpha A \alpha^{-1}$, and A is conjugate to B .

$\Leftarrow$: Suppose A is conjugate to B . So there is an isomorphism $\alpha : F^n \rightarrow F^n \ni B = \alpha A \alpha^{-1}$. Thus $\alpha \circ A = B \circ \alpha$. Thus $\alpha \circ A^k = B^k \circ \alpha$. Thus $\alpha \circ f(A) = f(B) \circ \alpha, \forall f \in F[X]$. Thus ϕ is an isomorphism of M_A and M_B since $M_A \cong F^n$ and $M_B \cong F^n$. □